How a Question Is Answered

Step 6 · `src/qa/` · `python -m src.qa "question" --user normal` · 2026-10-27

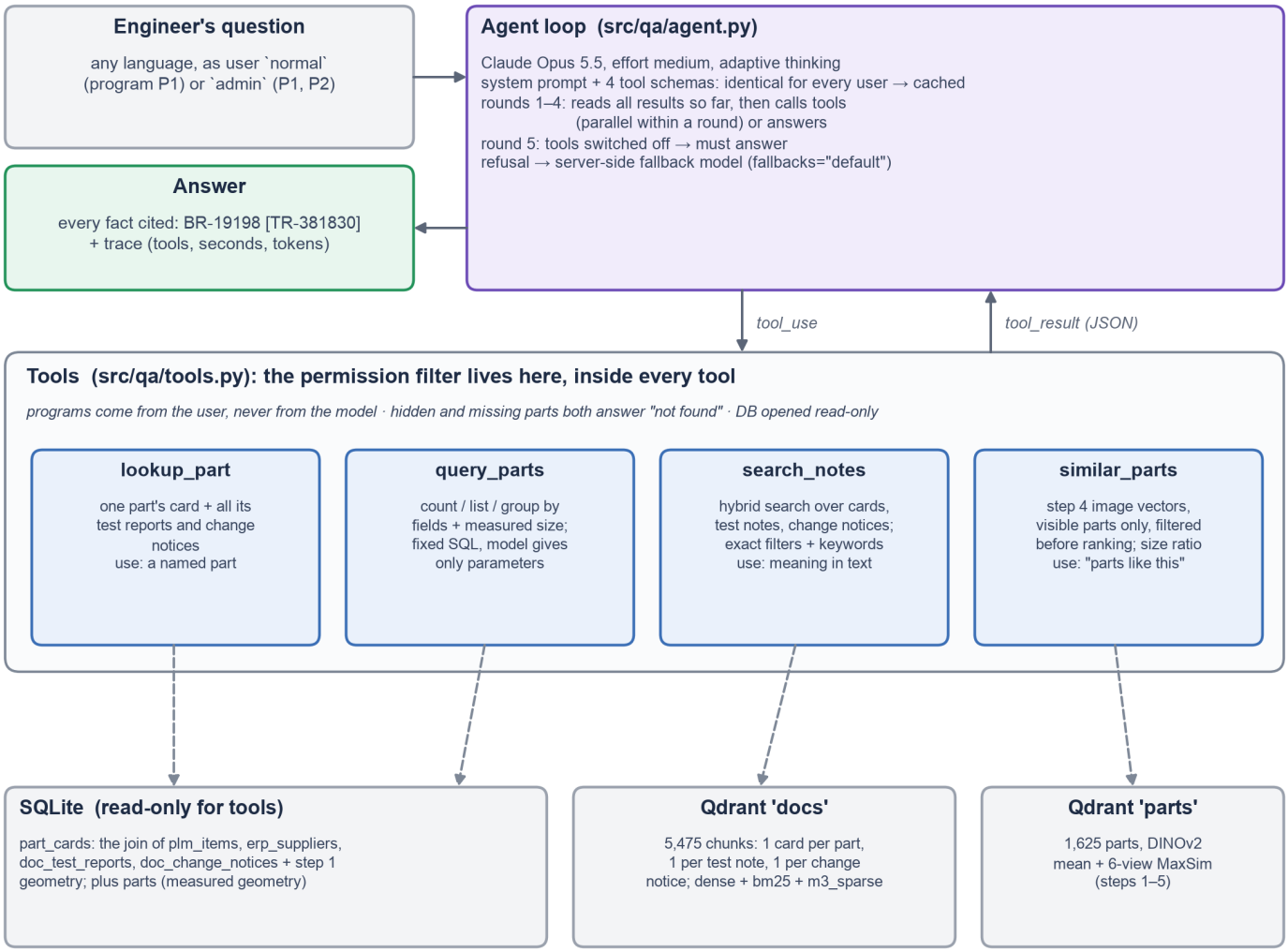


Figure 1. Claude chooses the tools; the tools decide what the user may see; the answer cites every fact.

Prompt rules

Hard facts go into filters (search can't see negation); bilingual synonyms as keywords; cite only ids a tool returned; name the revision a result belongs to; never guess.

Tools

Tool	Model controls	Permission
lookup_part	part number	hidden = missing → 'not found'
query_parts	parameters only; the SQL is fixed	program condition added by code
search_notes	query, keywords, exact filters	filter inside each prefetch
similar_parts	part number, size ratio	only visible parts are ranked

Data

Four scattered tables (plm_items, erp_suppliers, doc_test_reports, doc_change_notices; 39% of notes Korean) are joined into one part card per part. 5,475 chunks: cards + one per note; each note embedded with a header (part, class, material, test, result) that is never cited.

Retrieval

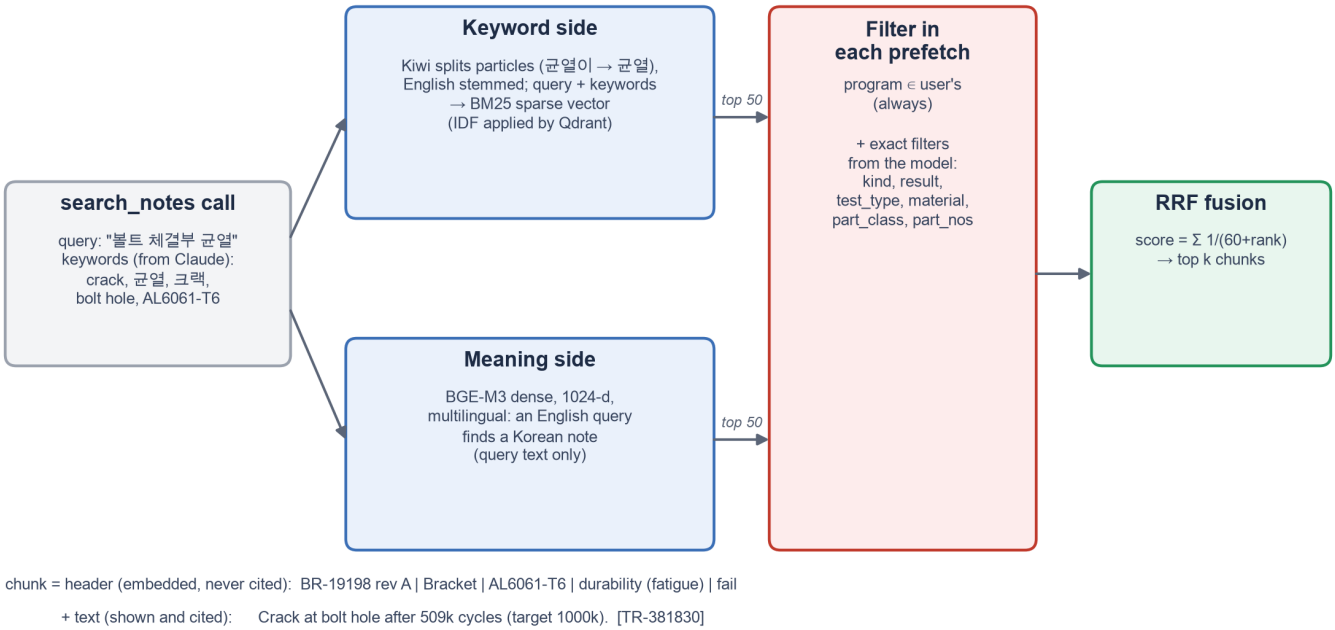


Figure 2. search_notes: keyword and meaning rankings, both restricted before ranking, merged by RRF.

Test	Result
크랙 난 브라켓 ('cracked bracket')	3 of top 5 said 'no crack'; fixed by result = fail filter
English query, BM25 / dense	BM25 matched wrong literal words; dense ignored the material
BM25 + bilingual expansion	4 of 4 correct, all from Korean notes

Measured

- Permissions: 14 vs 33 parts for normal vs admin; a direct question about a P2 part leaked nothing.
- 10–27 s per question once BGE-M3 is loaded (22.7 s, once); output tokens are the bottleneck.
- ~$0.07 per question; one 60-question evaluation pass ≈ $4; concurrent runs via `ask_many`.